

Demonstration of Proposed Features

Date	15 July 2026
Team ID	
Project Name	SkillEcho - Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S. No	Feature Name	Description	Status (Implemented/Partial/Pending)	Demonstrated (Yes/No)	Remarks
1	User Registration	Students and admins can create accounts with name, email, password, and role.	Implemented	Yes	Live registration on Streamlit UI -> POST /auth/register -> 201 response.
2	User Login	Authenticated access with SHA-256 password verification.	Implemented	Yes	Live login -> POST /auth/login -> user profile returned; role-based dashboard switch.
3	Concept Library	View all reference concepts with titles and descriptions.	Implemented	Yes	Concept dropdown populated from GET /concepts; 5 seeded concepts visible.
4	Concept PDF Viewer	View/download reference PDF for each concept.	Implemented	Yes	PDF download via GET /concepts/{id}/pdf in browser.
5	Admin Concept CRUD	Create, update, and delete concepts with optional PDF upload.	Implemented	Yes	Live creation of a 'Cloud Computing' concept with PDF, visible in student list; update and delete also shown.
6	Audio Upload	Students upload WAV audio recordings for evaluation.	Implemented	Yes	Live audio recording via Streamlit widget, uploaded to /evaluate.
7	Whisper Transcription	Audio transcribed to English text via Whisper STT.	Implemented	Yes	Transcript displayed on scorecard: clear, normalised English text from spoken input.
8	Topic Guardrail	Off-topic submissions detected and flagged with score 0.	Implemented	Yes	Off-topic audio against 'Machine Learning' concept returned 'Topic Mismatch', score = 0.
9	Acoustic Features	RMS energy, pause ratio, zero-crossing rate extracted.	Implemented	Yes	Signal metrics displayed on scorecard with numeric values.
10	Filler Word Analysis	Filler word count, ratio, and per-word breakdown.	Implemented	Yes	Filler breakdown shown (e.g. 'um': 3, 'like': 2, ratio: 4.2%).
11	Bi-Encoder Similarity	Semantic similarity score (0-100%) using MiniLM-L6-v2.	Implemented	Yes	Bi-Encoder % displayed on scorecard (e.g. 72.4%).
12	Cross-Encoder Accuracy	Semantic accuracy score (0-100%) using ms-marco Cross-Encoder.	Implemented	Yes	Cross-Encoder % displayed alongside Bi-Encoder on

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					scorecard.
13	Weighted Scoring	Composite 0-100 score with tier classification.	Implemented	Yes	Score 85 -> 'Strong Understanding' (green); Score 62 -> 'Moderate Understanding' (orange).
14	PDF Report Download	Professional evaluation report with metrics table and waveform.	Implemented	Yes	PDF downloaded via st.download_button() and opened to show the formatted report.
15	Evaluation History	Students view past evaluations with full metric details.	Implemented	Yes	History page shows ordered list of past evaluations with scores, concepts, and timestamps.
16	Admin Results Dashboard	Admins view all student evaluation results.	Implemented	Yes	Admin view shows all evaluations across users with name, email, and scores.
17	Admin User Management	Admins view all registered users.	Implemented	Yes	User list displayed with name, email, role, and registration date.
18	Health Check	API health endpoint.	Implemented	Yes	curl /health -> {"status": "ok", "version": "2.0.0"}.
19	API Documentation	Auto-generated Swagger UI.	Implemented	Yes	Swagger UI at /docs showing all 12 endpoints with schemas.
20	Cloud Deployment	Running on Streamlit Community Cloud.	Implemented	Yes	Live URL accessible from an external browser.

Feature Implementation Summary:

Metric	Value
Total Features Proposed	20
Total Features Implemented	20
Total Features Demonstrated	20
Overall Implementation Rate (%)	100%